



IE415 Control of Autonomous System

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Project Report

**State Feedback Control and Observer Design for
a 2 DOF Robotic Arm**

State Feedback Controller

State feedback control is a foundational concept in control theory and engineering, focusing on the control of a system by feeding back its state to determine an appropriate control action. This technique is widely used in various engineering applications to stabilize systems, improve their performance, and ensure desired responses. A **state feedback controller** uses the state of a system, represented as a vector of variables that describe the system's conditions (e.g., position, velocity, temperature), to compute control actions. In mathematical terms, this approach typically applies to **linear time-invariant systems** expressed in **state-space representation**, where the system dynamics can be described by matrices that relate input, state, and output.

State-Space Representation:

In the state-space approach, a dynamic system is typically represented by a set of first-order differential (or difference) equations:

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

where:

- $x(t)$ is the state vector, encapsulating the system's internal state variables.
- A is the state matrix that determines the system's internal dynamics.
- $u(t)$ is the control input vector applied to the system.
- B is the input matrix that maps the effect of the input to the state.
- $y(t)$ is the output vector, representing the measurable output variables of the system.
- C and D are output and feedthrough matrices, respectively.

The goal in state feedback control is to design the control input $u(t)$ so that the state vector $x(t)$ tracks a desired trajectory or reaches a specific point over time.

Control Law and Pole Placement:

One of the primary methods for designing a state feedback controller is **pole placement**. This involves adjusting the poles (eigenvalues of the closed-loop system matrix) to specific locations in the complex plane to achieve desired performance characteristics, like stability, faster response, or damping of oscillations. The state feedback control law is generally defined as:

$$u(t) = -Kx(t)$$

where K is the **state feedback gain matrix** designed to place the closed-loop poles at specific locations. By adjusting K , one can alter the dynamics of the closed-loop system (the system with feedback applied), such as by shifting its poles to the left half of the complex plane for stability or to fine-tune transient and steady-state performance.

Controller Design Process

The typical steps for designing a state feedback controller are:

1. **Define System Dynamics:** Represent the system in state-space form.
2. **Check Controllability:** Determine if the system is controllable, meaning all states can be manipulated by the input $u(t)$. A system is controllable if the **controllability matrix** (formed by matrices A and B) has full rank.
3. **Pole Placement or Optimal Control:** Based on the desired performance (e.g., response time, stability margin), select pole locations and compute the gain matrix K . Alternatively, methods like **Linear Quadratic Regulator (LQR)** can be used for optimal control.

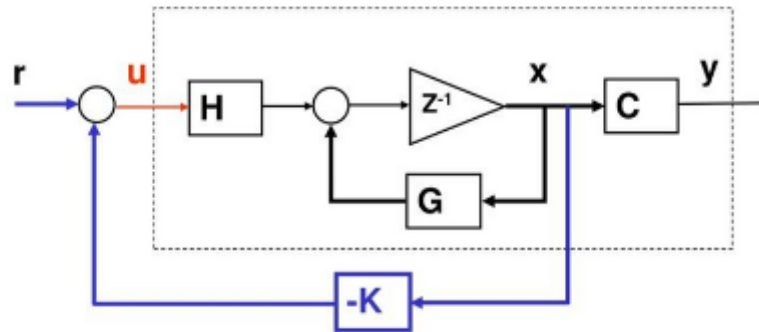
Tracking Control Theory:

In control theory, the **tracking problem** involves designing a control system that ensures the output of a system follows a desired reference trajectory or setpoint over time, despite disturbances or uncertainties in the system dynamics. The goal is to make the output $y(t)$ match the reference signal $r(t)$ as closely as possible.

Key Concepts in Tracking Control

1. **Error Dynamics:** The tracking error $e(t)$ is defined as the difference between the desired output $r(t)$ and the actual output $y(t)$, or $e(t) = r(t) - y(t)$. The objective is to minimize $e(t)$ over time.
2. **Reference Model and Control Law:** In state feedback tracking, a control law $u(t) = -Kx(t) + Nr(t)$ is used, where K is the feedback gain and N is a feedforward gain to ensure steady-state tracking of $r(t)$. The design of K and N depends on desired performance metrics, such as settling time and overshoot.
3. **Stability and Convergence:** A well-designed tracking controller ensures that $e(t)$ approaches zero as t increases, achieving accurate tracking. Techniques like **pole placement** or **optimal control** (e.g., LQR) help determine the necessary gains to ensure stability and meet performance requirements.

State Feedback Control Block Diagram



$$u = -Kx + r$$

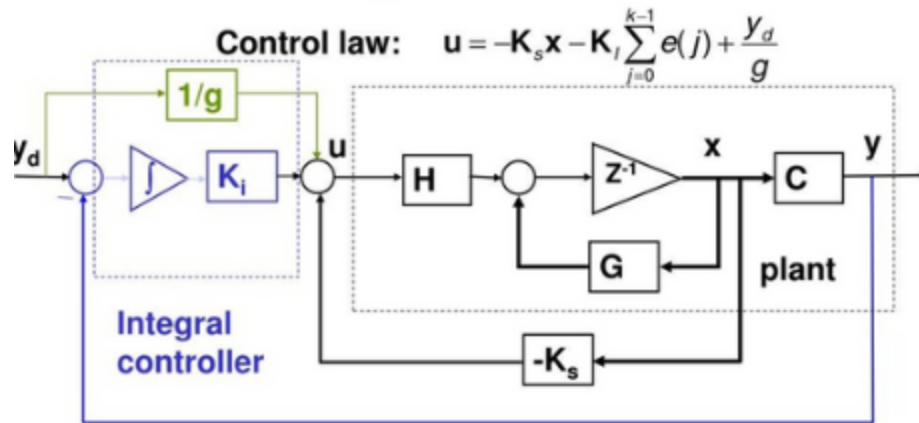
Closed-loop System

Plant: $x(k+1) = Gx(k) + Hu(k)$
 $y(k) = Cx(k)$

Control: $u = -Kx + r$

Closed-loop System: $x(k+1) = (G - HK)x(k) + Hr(k)$
 $y = Cx$

Integral Control



Automatically generates reference input r !

Controllability

Controllability refers to the ability to drive the state of a system from any initial state to any desired final state within a finite time by applying an appropriate control input. If a system is controllable, it means that each state variable can be influenced by the control input, making it possible to stabilize or modify the system as needed.

The system is described by: $\dot{x} = Ax + Bu$ and the control law is defined as:

$$u = -Kx$$

To check controllability, compute the controllability matrix M or $Q_{sc,n}$:

$$Q_{sc,n} = M = [B \ AB \ A^2B \ \dots \ A^{n-1}B]. \text{ If full rank, the system is state controllable.}$$

$$K = [\alpha_n - a_n \ \alpha_{n-1} - a_{n-1} \ \dots \ \alpha_2 - a_2 \ \alpha_1 - a_1] T^{-1}$$

This gives the feedback gain matrix K , which forces the eigenvalues of the closed-loop system to be the desired values $\mu_1, \mu_2, \dots, \mu_n$.

Importance of Controllability

If a system is controllable, a state feedback controller can be designed to modify the system dynamics, such as by placing the poles in desired locations for stability or improving response characteristics. If a system is not controllable, certain states cannot be influenced by any input, which limits the control over the system.

Observability

Observability refers to the ability to reconstruct the state of a system by observing its outputs over time. If a system is observable, it means that the entire internal state can be inferred from the output measurements, which is essential for designing state observers.

For an LTI system represented as:

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

where $y(t)$ is the output vector, and C is the output matrix, the system is **observable** if the

observability matrix $\mathcal{O} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-1} \end{bmatrix}$ has full rank. Full rank here indicates that all state variables

can be inferred from the output measurements.

Importance of Observability

When a system is observable, an observer (or estimator) can be designed to estimate the state vector $x(t)$ from measurements of $y(t)$. This is particularly valuable in cases where not all states are directly measurable, enabling state feedback control even in the absence of complete state information.

Application of Control Theories in the 2 DOF Arm Robot Project

Designing a 2 DOF arm robot involves addressing various challenges in tracking, stabilization, and precise movement. The system's control strategy must account for factors such as stability, accuracy, and responsiveness, which can be achieved through a combination of **system feedback**, **PID control**, and **state-space methods** involving **controllability** and **observability**. Each of these theories plays a critical role in designing an effective control system for the robot's two joints (shoulder and elbow), enabling it to reach target positions and maintain desired trajectories accurately.

1. System Feedback Theory in Robot Control

System feedback refers to the use of information about a system's current state or output to adjust the control inputs in real-time. For a 2 DOF robotic arm, the state variables typically include the joint angles and their velocities. By applying feedback, we can dynamically adjust the control signals to correct deviations from the desired state or trajectory.

In this project, system feedback is crucial for:

- **Position Control:** Feedback ensures that the end effector of the arm reaches the specified position accurately.
- **Stabilization:** Real-time feedback enables the system to counteract disturbances or changes in load (e.g., when the robot is holding an object).

The **state feedback control law** is commonly used here, where the control input $u(t)$ is calculated as:

$$u(t) = -Kx(t)$$

with K being the gain matrix that determines how aggressively the controller responds to deviations. This feedback law can adjust the robot's joint torques, allowing for precise positioning and smooth movements, critical for applications requiring accuracy.

2. PID Controller in Position and Speed Control

The **PID (Proportional-Integral-Derivative) controller** is one of the most widely used control methods in robotic systems due to its simplicity and effectiveness in managing both position and speed control. In a 2 DOF arm robot, each joint can have its own PID controller to independently regulate the angular position and velocity, which allows the robot to maintain accurate trajectories.

PID Control Components

- **Proportional Control (P):** This component applies control based on the present error. In the 2 DOF arm robot, proportional control provides an immediate response to position or velocity errors in the joints, helping the robot move quickly towards the target.
- **Integral Control (I):** The integral term accumulates past errors to correct any steady-state error over time, ensuring the arm reaches and stays at the target position without residual error.
- **Derivative Control (D):** The derivative term anticipates future errors by responding to the rate of change of the error, helping to smooth out oscillations and ensuring a stable response.

The combined PID control law can be written as:

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

For the 2 DOF arm robot, PID controllers allow smooth and accurate control of joint angles, which is essential for maintaining a desired path. Additionally, each joint's PID controller can be tuned to address specific response characteristics, such as reducing overshoot or increasing responsiveness.

3. Controllability and Observability in State-Space Design

In more complex robotic systems like a 2 DOF arm robot, **state-space representation** provides a structured approach for modeling and controlling the system. This approach is essential because it allows us to design controllers that are not only effective but also optimal in handling dynamic behaviors.

Controllability in the 2 DOF Arm Robot

Controllability ensures that each joint's position and velocity can be influenced by the control inputs. For the 2 DOF robot, the **controllability matrix** must have full rank, meaning that it can control both states—joint angles and their velocities—across the two degrees of freedom. With controllability, we can design a feedback controller that drives the robot's joints to the desired position and trajectory.

If the robot is not fully controllable, certain movements or adjustments cannot be achieved, limiting the robot's functionality. For example, if the system lacks controllability, one or both of the joints may fail to reach certain positions due to a lack of control influence. Therefore, controllability is a foundational requirement for the state feedback control design in the robot.

Observability in the 2 DOF Arm Robot

Observability, on the other hand, ensures that the internal states (such as angles and velocities) can be reconstructed from the output measurements. In the 2 DOF arm robot, observability means that by measuring each joint's position and velocity, we can determine the full state of the robot.

For practical purposes, it's often challenging to directly measure all internal states, especially in larger or more complex robotic systems. Therefore, the use of **observers** (such as a **Luenberger Observer**) becomes critical. An observer uses the available measurements to estimate unmeasured states, ensuring that the controller receives accurate state information for feedback purposes.

In this project, a **state observer** can help estimate velocities when only positions are measured. The controller can then use this information to adjust the joint torques accurately, ensuring smooth and stable motion of the robotic arm.

Integrating Control Theories for Effective 2 DOF Arm Robot Design

By combining system feedback, PID control, and state-space methods, we can create a robust control framework for the 2 DOF arm robot:

- **System Feedback** provides real-time adjustments, helping to achieve stability and precise positioning.
- **PID Control** in each joint allows for independent and smooth control of both angular position and velocity.
- **State-Space Representation** along with controllability and observability ensures that the entire system is fully controllable and that we can infer the necessary states, enabling accurate and optimal control.

Now, given the joint angles θ_1 , θ_2 we can determine the end effector coordinates x and y .

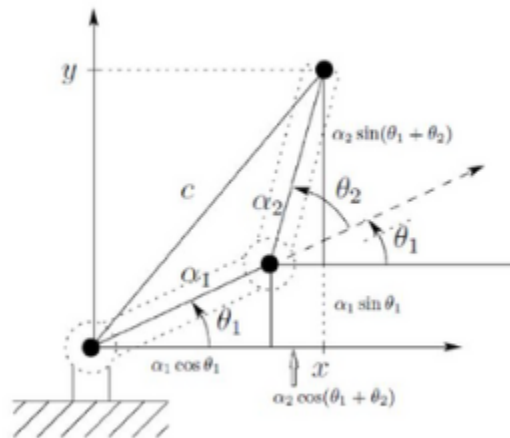


Fig. 1.26 Solving for the joint angles of a two-link planar arm.

Forward kinematics

The coordinates (x, y) of the tool are expressed in this coordinate frame as:

$$x = x_2 = \alpha_1 \cos \theta_1 + \alpha_2 \cos(\theta_1 + \theta_2) \quad (1.1)$$

$$y = y_2 = \alpha_1 \sin \theta_1 + \alpha_2 \sin(\theta_1 + \theta_2) \quad (1.2)$$

in which α_1 and α_2 are the lengths of the two links, respectively. Also the orientation of the tool frame relative to the base frame is given by the direction cosines of the x_2 and y_2 axes relative to the x_0 and y_0 axes, that is,

$$\begin{aligned} x_2 \cdot x_0 &= \cos(\theta_1 + \theta_2); & x_2 \cdot y_0 &= -\sin(\theta_1 + \theta_2) \\ y_2 \cdot x_0 &= \sin(\theta_1 + \theta_2); & y_2 \cdot y_0 &= \cos(\theta_1 + \theta_2) \end{aligned}$$

which we may combine into an orientation matrix

$$\begin{bmatrix} x_2 \cdot x_0 & y_2 \cdot x_0 \\ x_2 \cdot y_0 & y_2 \cdot y_0 \end{bmatrix} = \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) \end{bmatrix} \quad (1.3)$$

Inverse Kinematics

In order to command the robot to move to a location we need the inverse; that is, we need the joint variables θ_1, θ_2 in terms of the x and y coordinates of A.

This is the problem of inverse kinematics. In other words, given x and y in the forward kinematic Equations (1.1) and (1.2).

Consider the diagram of Figure 1.26. Using the Law of Cosines we see that the angle θ_2 is given by

$$\cos \theta_2 = \frac{x^2 + y^2 - \alpha_1^2 - \alpha_2^2}{2\alpha_1\alpha_2} := D \quad (1.4)$$

We could now determine θ_2 as

$$\theta_2 = \cos^{-1}(D) \quad (1.5)$$

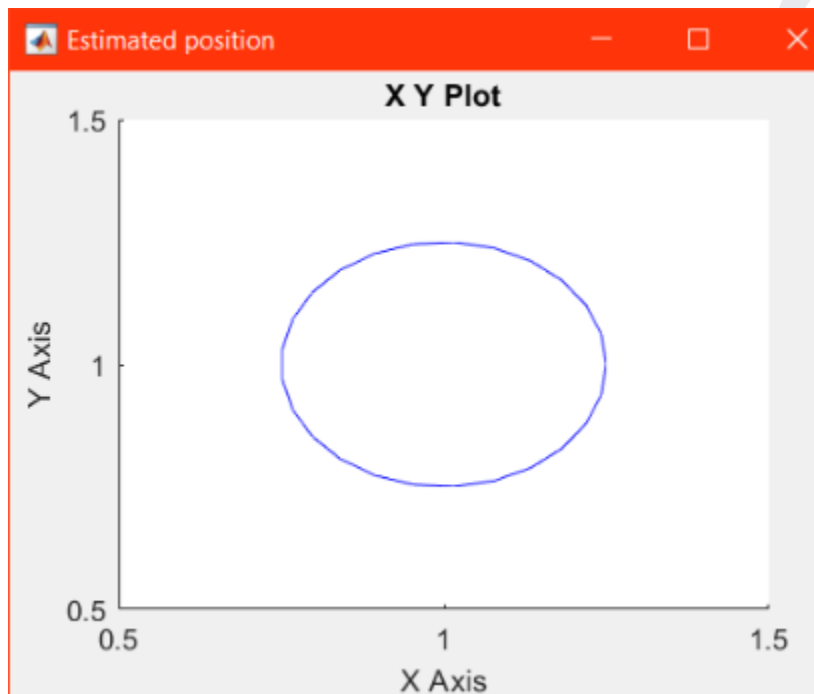
However, a better way to find θ_2 is to notice that if $\cos(\theta_2)$ is given by Equation (1.4) then $\sin(\theta_2)$ is given as

$$\sin(\theta_2) = \pm\sqrt{1 - D^2} \quad (1.6)$$

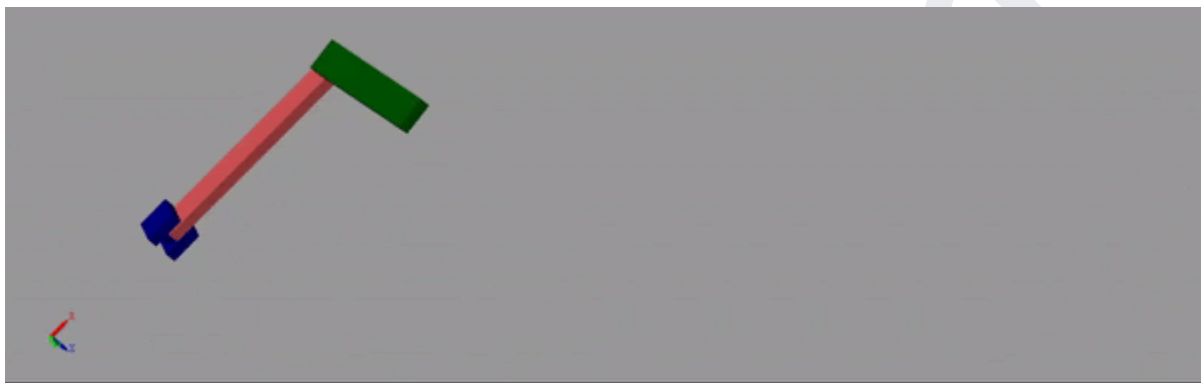
and, hence, θ_2 can be found by

$$\theta_2 = \tan^{-1} \frac{\pm\sqrt{1 - D^2}}{D} \quad (1.7)$$

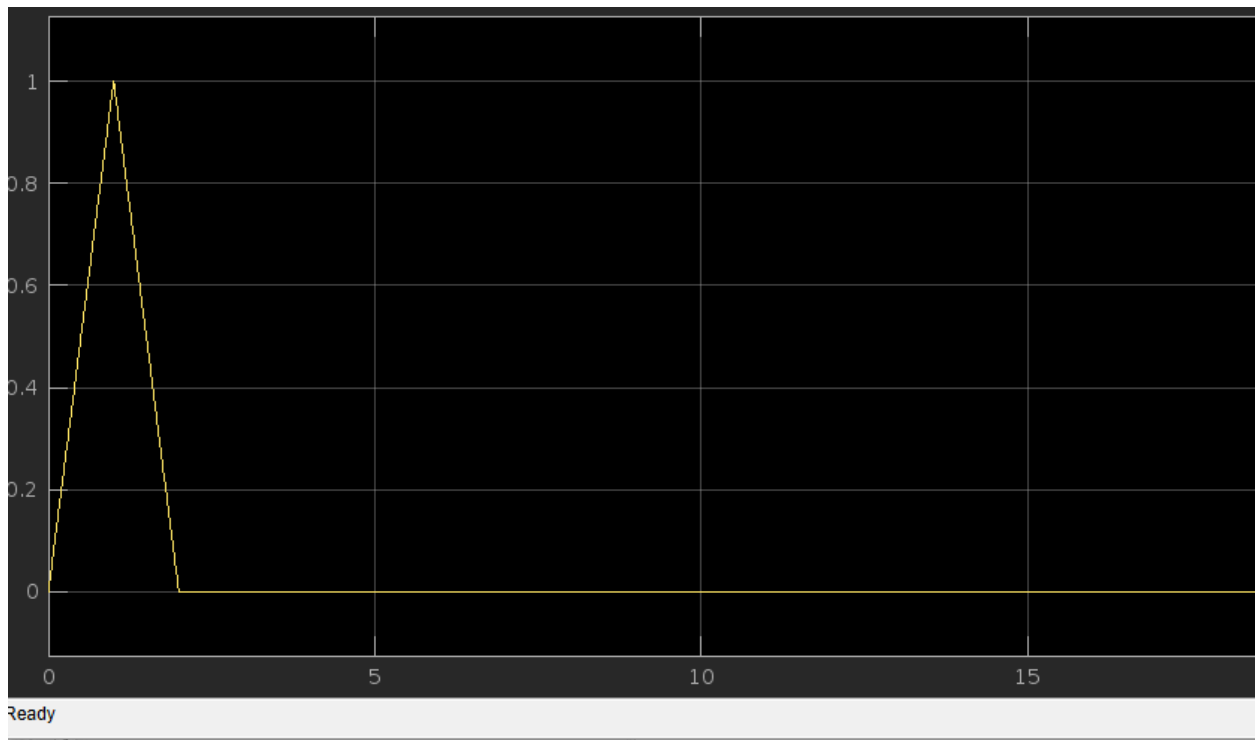
The image below illustrates the expected path of a 2 DOF robotic arm in a 2D plane. This path represents the target trajectory that the robot's end effector (the tip of the arm) should follow, with precise control of its two joints (shoulder and elbow). The control system, incorporating feedback and trajectory tracking, is designed to enable the arm to reach each position along this path smoothly and accurately, simulating a controlled and responsive movement within the robot's operational range.



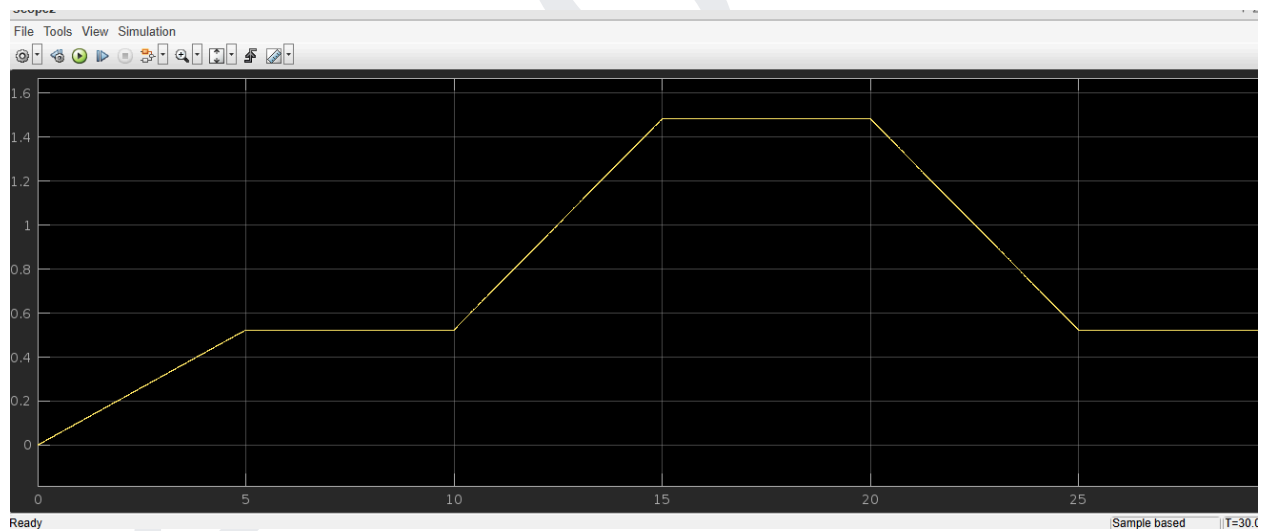
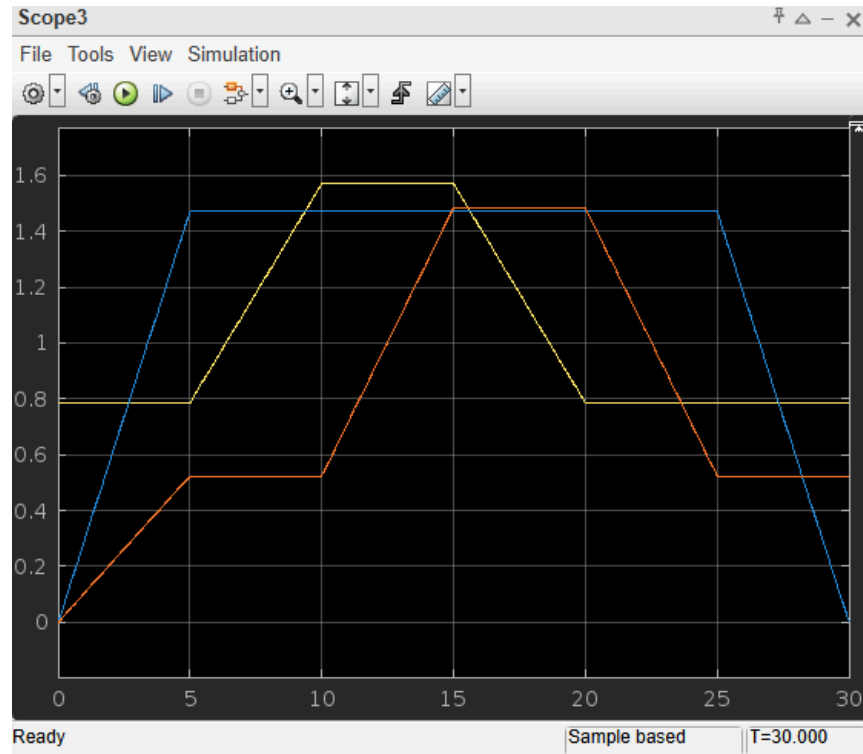
The Video below displays the simulation results of the 2 DOF robotic arm in Simulink, showing how effectively the arm follows its intended trajectory in a 2D plane. These results demonstrate the performance of the control system, with real-time adjustments in joint positions to accurately track the desired path. The simulated response validates the effectiveness of the feedback control and trajectory tracking mechanisms, ensuring smooth and precise movement of the arm's end effector along the specified path, even under dynamic conditions.



- The Scope Output of the Simscape Matlab:



→ This shows a signal that spikes initially and then settles to zero, which represent an impulse response or a damped system reaching equilibrium (possibly acceleration or force on one DOF).



→ This graph shows a single signal rising, holding a plateau, and then decreasing symmetrically, which is characteristic of a displacement or position signal that follows a trajectory over time.